

Section D — AI-Augmented Learning

Cisco Packet Tracer — Router-on-a-Stick Practical

STEP 1 • BUILD WITH AI

AI Tool Used: ChatGPT

1. Exact Prompt Given to AI

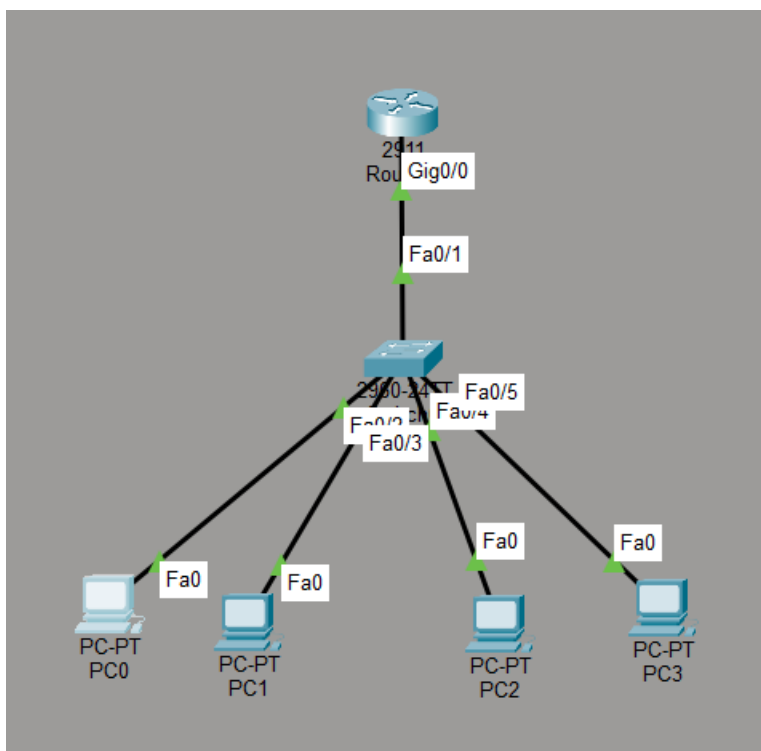
I am doing a Cisco Packet Tracer practical. Generate a complete Cisco IOS configuration for a Router-on-a-Stick topology.

Requirements:

- 1. Configure VLAN 10 and VLAN 20 on the switch.*
 - 2. Assign switch ports to VLAN 10 and VLAN 20.*
 - 3. Configure an 802.1Q trunk between the switch and router.*
 - 4. Configure Router-on-a-Stick sub-interfaces on the router for VLAN 10 and VLAN 20.*
 - 5. Configure IP addresses/gateways so that a PC in VLAN 10 can communicate with a PC in VLAN 20.*
 - 6. Provide complete switch and router commands.*
 - 7. Explain how to configure the PCs.*
 - 8. The final configuration should allow successful ping between the VLAN 10 and VLAN 20 hosts.*
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2. AI Original Configuration Script

Topology:



IP Address Plan:

Device	VLAN	IP Address	Subnet Mask	Gateway
PC0	10	192.168.10.10	255.255.255.0	192.168.10.1
PC1	20	192.168.20.10	255.255.255.0	192.168.20.1
Router Sub-interface	10	192.168.10.1	255.255.255.0	—
Router Sub-interface	20	192.168.20.1	255.255.255.0	—

Switch — AI Original:

```
enable
configure terminal

vlan 10
  name VLAN10
exit

vlan 20
  name VLAN20
exit

interface fastEthernet 0/1
  switchport mode access
  switchport access vlan 10
  no shutdown
exit

interface fastEthernet 0/2
  switchport mode access
  switchport access vlan 20
  no shutdown
exit

interface gigabitEthernet 0/1
  switchport mode trunk
  no shutdown
exit

end
write memory
```

Router — AI Original:

```
enable
configure terminal

interface gigabitEthernet 0/0
  no shutdown
exit

interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
exit

end
write memory
```

3. Testing in Packet Tracer

PC0 Configuration:

IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

PC1 Configuration:

IP Address: 192.168.20.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.20.1

Test command from PC0:

ping 192.168.20.10

STEP 2 · TEST & DEBUG (WITHOUT AI)

Example bug found during testing: the trunk must carry both VLAN 10 and VLAN 20. If the trunk is restricted to only VLAN 10, VLAN 20 traffic cannot cross the switch-router trunk.

Corrected Switch Trunk Configuration:

```
interface gigabitEthernet 0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
exit
```

Corrected Version — Complete Switch Configuration

```
enable
configure terminal

vlan 10
  name VLAN10
exit

vlan 20
  name VLAN20
exit

interface fastEthernet 0/1
  switchport mode access
  switchport access vlan 10
  no shutdown
exit

interface fastEthernet 0/2
  switchport mode access
  switchport access vlan 20
  no shutdown
exit

interface gigabitEthernet 0/1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
exit

end
write memory
```

Corrected Version — Router Configuration

```
enable
configure terminal
```

```
interface gigabitEthernet 0/0
  no shutdown
exit

interface gigabitEthernet 0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
exit

end
write memory
```

3–4 Line Debugging Note

I tested the AI-generated configuration in Cisco Packet Tracer and found that the trunk needed to carry both VLAN 10 and VLAN 20. I corrected the trunk configuration by explicitly allowing VLANs 10 and 20. After making the correction, the Router-on-a-Stick configuration worked correctly and PC0 in VLAN 10 successfully pinged PC1 in VLAN 20. This was necessary because inter-VLAN traffic must be carried through the trunk for both VLANs.